

with the contours $\Gamma_{\pm}$. It is found that, when grade restricting the branes of Eq. (4.7) to a specific window, the open Witten index was consistent when calculated on the positive phase and on the negative phase, *i.e.* $\chi_{k,N}^+ = \chi_{k,N}^-$, suggesting strongly that successful brane transport has taken place. The results of a few examples are presented below.

For $K_{Gr(2,4)}$, the basis of branes used was $\{(0,0), (1,0), (1,1), (2,0), (2,1), (2,2)\}$ and the open Witten indices was calculated to be

$$\chi_{2,4}^+ = \chi_{2,4}^- = \begin{pmatrix} 0 & 4 & 6 & 10 & 20 & 20 \\ -4 & 0 & 4 & 4 & 16 & 20 \\ -6 & -4 & 0 & 0 & 4 & 6 \\ -10 & -4 & 0 & 0 & 4 & 10 \\ -20 & -16 & -4 & -4 & 0 & 4 \\ -20 & -20 & -6 & -10 & -4 & 0 \end{pmatrix}. \quad (4.14)$$

A special note is needed here: this was found to be consistent for a set of branes all grade restricted to one of the charge window types in Fig. 4. On the other hand, if a set of branes grade restricted to different window types were mixed together, so that the entire set was not grade restricted to the same charge window type, then the open Witten indices was found to be inconsistent. Thus it is believed that branes must be grade restricted in exactly the same way when computing the Witten index, despite multiple ways of charge restriction being valid for a single choice of theta-angle.

For $K_{Gr(2,5)}$, the basis of branes used was $\{(0,0), (1,0), (1,1), (2,0), (2,1), (2,2), (3,0), (3,1), (3,2), (3,3)\}$. Then we found¹⁴

$$\chi_{2,5}^+ = \chi_{2,5}^- = \begin{pmatrix} 0 & -5 & -10 & -15 & -40 & -50 & -35 & -105 & -175 & -175 \\ 5 & 0 & -5 & -5 & -25 & -40 & -15 & -75 & -155 & -175 \\ 10 & 5 & 0 & 0 & -5 & -10 & 0 & -15 & -40 & -50 \\ 15 & 5 & 0 & 0 & -5 & -15 & -5 & -25 & -75 & -105 \\ 40 & 25 & 5 & 5 & 0 & -5 & 0 & -5 & -25 & -40 \\ 50 & 40 & 10 & 15 & 5 & 0 & 0 & 0 & -5 & -10 \\ 35 & 15 & 0 & 5 & 0 & 0 & 0 & -5 & -15 & -35 \\ 105 & 75 & 15 & 25 & 5 & 0 & 5 & 0 & -5 & -15 \\ 175 & 155 & 40 & 75 & 25 & 5 & 15 & 5 & 0 & -5 \\ 175 & 175 & 50 & 105 & 40 & 10 & 35 & 15 & 5 & 0 \end{pmatrix}. \quad (4.15)$$

For $K_{Gr(2,6)}$, we have 15 generators so, we just present the computation for the subset $\{(0,0), (1,0), (1,1), (2,1), (2,2), (3,2), (3,3), (4,3), (4,4)\}$:

$$\chi_{2,6}^+ = \chi_{2,6}^- = \begin{pmatrix} 0 & 6 & 15 & 70 & 105 & 420 & 490 & 1764 & 1764 \\ -6 & 0 & 6 & 36 & 70 & 315 & 420 & 1624 & 1764 \\ -15 & -6 & 0 & 6 & 15 & 70 & 105 & 420 & 490 \\ -70 & -36 & -6 & 0 & 6 & 36 & 70 & 315 & 420 \\ -105 & -70 & -15 & -6 & 0 & 6 & 15 & 70 & 105 \\ -420 & -315 & -70 & -36 & -6 & 0 & 6 & 36 & 70 \\ -490 & -420 & -105 & -70 & -15 & -6 & 0 & 6 & 15 \\ -1764 & -1624 & -420 & -315 & -70 & -36 & -6 & 0 & 6 \\ -1764 & -1764 & -490 & -420 & -105 & -70 & -15 & -6 & 0 \end{pmatrix}. \quad (4.16)$$

¹⁴We remark that (4.15) was computed in [15]. It corresponds to the open Witten index computed in what is called the 'S-basis' in [15]. More precisely, using the notation of [15]: $(\chi_-^{-T} - \chi_-^{-1}) = \chi_{2,5}^+ = \chi_{2,5}^-$.